QEF: corrected Table 7 and source conditions

Original three-layer operators; all-model reference calculations

Numerical source certification is not established by the original generator specification.

1 Security model and a rotation-moment bound

The security quantity considered here is the number of independently encrypted quantum preparations needed to distinguish two specified images. This is a sample-complexity statement, rather than a gate-complexity estimate for exhaustive key search. The proof uses the original three QEF layers. It does not assume that an adversary reconstructs an arbitrary unitary or recovers every component of the original key.

Keys, ciphertexts, and the attack experiment. Write A for the substitution-angle material produced by the nD-QHCS and $B = (K^{\mathrm{perm}}, K^{\mathrm{ent}})$ for the material produced by the HGΦ-QRNG. These are two source classes. The two components of B are described by their joint distribution; independence is not inferred from their output lengths. Each encryption uses a fresh secret key record, and the receiver obtains the same record over an assumed secure channel. Independent refresh between preparations is an explicit security assumption, stronger than merely obtaining different numerical key values. The experiment gives the adversary no additional side information correlated with the target secret record; any claimed extension to source leakage must bound that leakage explicitly.

Let $N = HW$, let r denote the feature-register size, and let $D = N2^r$ denote the dimension of the valid image subspace. Unused padded position addresses are excluded from the ciphertext experiment. The image states have the form

$$|I\rangle = N^{-1/2} \sum_{k=0}^{N-1} |k\rangle|\chi_k\rangle, \qquad \langle\chi_k|\chi_k\rangle = 1. \tag{1}$$

The adversary is given the classical descriptions of two such images of the same public dimensions. A uniform hidden bit b selects I_b. The adversary receives q encryptions of that selected image, with independently refreshed keys, and may perform any joint quantum measurement. The publicly transmitted ciphertext is the quantum state, not a classical list of all its amplitudes or a rendered image file. Define

$$\sigma_b = \mathbb{E}_{A,B}[U_{A,B}|I_b\rangle\langle I_b|U_{A,B}^\dagger]. \tag{2}$$

The joint ciphertext is $\sigma_b^{\otimes q}$. Its optimal binary discrimination probability is [1]

$$P_{\mathrm{guess}}^{(q)} = \frac{1}{2} + \frac{1}{4}\left\|\sigma_0^{\otimes q} - \sigma_1^{\otimes q}\right\|_1. \tag{3}$$

This experiment does not give repeated access to one fixed-key unitary. It also does not cover key reuse, arbitrary reference-entangled chosen inputs, authentication, or publication of simulator amplitude records.

Use the actual circuit structure. For the intended pixel-dependent substitution, write

$$R_A = \sum_k |k\rangle\langle k| \otimes R_{A,k}, \qquad R_{A,k} = \bigotimes_{j=1}^{r} T_{k,j}, \quad T_{k,j} = R_z(\theta_{k,j}^z)R_y(\theta_{k,j}^y)R_x(\theta_{k,j}^x). \tag{4}$$

Position-controlled CNOT/CZ gates leave each position basis label unchanged. Thus, for suitable feature unitaries $V_B(t)$,

$$E_B = \sum_t |t\rangle\langle t| \otimes V_B(t), \qquad U_{A,B} = E_B P_B R_A = \sum_k |\pi_B(k)\rangle\langle k| \otimes V_B(\pi_B(k))R_{A,k}. \tag{5}$$

The permutation and entanglement keys may have any joint distribution in the theorem below. In particular, a uniform distribution over all $N!$ permutations is not needed for the image experiment in Eq. (1). The permutation must nevertheless be implemented reversibly with its work registers correctly cleaned.

An explicit angle reference law. The following law is an analytical reference, not an experimentally certified property of the nD-QHCS. For an integer $L \geq 2$, let each Euler angle be uniform on

$$\Theta_L = \{2\pi s/L : s = 0, \ldots, L-1\}. \tag{6}$$

Conditioned on B, the $3r$ angles within a pixel are independent, and the angle tuples belonging to any two different pixels are independent. Pairwise independence of pixel tuples is sufficient; independence of all N tuples jointly is not required. The continuous reference is the corresponding independent uniform law on $[0, 2\pi)$. These conditions must hold after conditioning on any public setup information or side information used in the attack experiment. The reference is a key-law assumption, not a consequence of deterministic seed expansion. For $N \geq 2$, two independent uniform angle tuples alone require at least $6r \log_2 L$ bits of conditional source entropy. The continuous law is an ideal limiting model; a finite implementation must use the finite-law expression and provide sufficient actual key material.

Set

$$a_L = \frac{1}{L \sin(\pi/(2L))}, \qquad \beta_L = a_L^{6r}, \qquad \beta_\infty = \left(\frac{2}{\pi}\right)^{6r}. \tag{7}$$

Uniform Euler coordinates are not Haar-distributed rotations. The nonzero first moment is retained in the analysis below.

Theorem 1 (Conditional image-discrimination sample bound)**.** *Under the stated angle reference law and independent key refresh, let $N \geq 2$, $F = 2^r$, $c_N = 1 - 1/N$, and*

$$\begin{aligned} x_+ &= 1 + F\beta_L(N-1), \qquad x_- = 1 - F\beta_L, \\ C_{N,r,L} &= \frac{x_+ \ln x_+ + (N-1)x_- \ln x_-}{NF}. \end{aligned} \tag{8}$$

Here $x_- > 0$ for the stated finite and continuous reference laws. For every image state in Eq. (1)*, its average ciphertext σ satisfies*

$$\frac{1}{2}\|\sigma - I_D/D\|_1 \leq \beta_L c_N, \tag{9}$$

$$\mathsf{D}(\sigma \| I_D/D) \leq C_{N,r,L}, \tag{10}$$

where D *uses natural logarithms. For any two such images, achieving $P_{\text{guess}}^{(q)} \geq p > 1/2$ requires*

$$\boxed{q \;\geq\; \left\lceil \max\left\{ \frac{\ln 2 - h(p)}{C_{N,r,L}},\; \frac{p - 1/2}{\beta_L(1 - 1/N)} \right\} \right\rceil}, \qquad h(p) = -p\ln p - (1-p)\ln(1-p). \tag{11}$$

The continuous reference uses β_∞ in the same expressions. For $N = 1$, the feature twirl makes the reference ciphertext exactly maximally mixed, and no finite number of fresh encryptions distinguishes the candidate images.

Proof. For one rotation axis a, averaging its conjugation over Θ_L dephases the Bloch components transverse to that axis. Composing the independent x, y, and z averages therefore gives

$$\mathbb{E}[TXT^\dagger] = \frac{I_2}{2} \operatorname{Tr} X. \tag{12}$$

For the first moment, a finite trigonometric sum gives

$$\mathbb{E}[R_a(\theta)] = c_L I_2 - i s_L \sigma_a, \quad c_L = 1/L, \quad s_L = L^{-1}\cot(\pi/(2L)). \tag{13}$$

Each matrix on the right is a_L times a unitary, since $c_L^2 + s_L^2 = a_L^2$. Hence $M = \mathbb{E}[R_{A,k}] = \beta_L^{1/2} W$ for a feature unitary W, and the feature conjugation average is the completely depolarizing channel. In the continuous limit,

$$\mathbb{E}[T] = (2/\pi)^3 I_2, \qquad M = (2/\pi)^{3r} I_{2^r}. \tag{14}$$

For example, the first identity follows by multiplying $\mathbb{E}[R_a] = -i(2/\pi)\sigma_a$ in the order z, y, x.

Write $\rho = |I\rangle\langle I| = \sum_{k,l} |k\rangle\langle l| \otimes X_{kl}$, and let Δ remove its off-diagonal position blocks. Conditional pairwise independence gives

$$\mathbb{E}[R_{A,k} X_{kl} R_{A,l}^\dagger] = \begin{cases} I_{2^r} \operatorname{Tr}(X_{kk})/2^r, & k = l, \\ M X_{kl} M^\dagger, & k \neq l. \end{cases} \tag{15}$$

For Eq. (1), $\operatorname{Tr} X_{kk} = 1/N$. Consequently the diagonal contribution is $\tau = I_D/D$, which is invariant under every $E_B P_B$. The remaining term is

$$\sigma - \tau = \beta_L \, \mathbb{E}_B[F_B(\rho - \Delta\rho)F_B^\dagger], \quad F_B = E_B P_B (I \otimes W). \tag{16}$$

Every F_B is unitary, so both trace norm and Hilbert–Schmidt norm are bounded by the corresponding norms before this average.

The pinching map is an average over N diagonal position-phase unitaries, one of which is the identity. The triangle inequality gives $\|\rho - \Delta\rho\|_1 \leq 2(1 - 1/N)$. Combining this with Eq. (16) proves Eq. (9). Moreover, because ρ is pure and its position probabilities are uniform,

$$\|\rho - \Delta\rho\|_2^2 = \operatorname{Tr}\rho^2 - \operatorname{Tr}(\Delta\rho)^2 = 1 - 1/N. \tag{17}$$

Thus $\|\sigma - \tau\|_2^2 \leq \beta_L^2 c_N$. A sharper entropy bound uses the exact spectrum before the final average. The N vectors $|k\rangle|\chi_k\rangle$ are orthonormal, and on their span $\rho - \Delta\rho$ has eigenvalues $1 - 1/N$ once and $-1/N$ with multiplicity $N - 1$. It is zero on the orthogonal complement. Consequently $\tau + \beta_L(\rho - \Delta\rho)$ has eigenvalues

$$\lambda_+ = x_+/D, \qquad \lambda_- = x_-/D \quad (N - 1 \text{ times}), \qquad \lambda_0 = 1/D \quad (N(F - 1) \text{ times}). \tag{18}$$

Every conditional output in Eq. (16) is a unitary conjugate of this state. Relative entropy is convex and τ is invariant, so

$$\mathsf{D}(\sigma\|\tau) \leq \lambda_+ \ln(D\lambda_+) + (N - 1)\lambda_- \ln(D\lambda_-) = C_{N,r,L}. \tag{19}$$

For comparison, Jensen's inequality gives the valid, weaker bound

$$C_{N,r,L} \leq \ln(1 + D\beta_L^2 c_N). \tag{20}$$

For a uniform hidden bit and q independent encryptions, the Holevo information obeys

$$\chi_q \leq \frac{1}{2} \sum_{b=0}^{1} \mathsf{D}(\sigma_b^{\otimes q} \| \tau^{\otimes q}) \leq q C_{N,r,L}. \tag{21}$$

Indeed, subtracting χ_q from the first expression gives the nonnegative relative entropy of the average ciphertext to $\tau^{\otimes q}$. Holevo's theorem and binary Fano's inequality [1] imply that any guess with success at least p requires mutual information at least $\ln 2 - h(p)$. Therefore $\ln 2 - h(p) \leq q C_{N,r,L}$. Independently, the trace-distance triangle inequality and Eq. (9) give $\frac{1}{2}\|\sigma_0 - \sigma_1\|_1 \leq 2\beta_L c_N$. Expanding the difference of the tensor powers as a telescoping sum, and using $\|\sigma_b\|_1 = 1$, yields

$$\frac{1}{2}\|\sigma_0^{\otimes q} - \sigma_1^{\otimes q}\|_1 \leq 2q\beta_L c_N, \qquad P_{\text{guess}}^{(q)} \leq \frac{1}{2} + q\beta_L c_N. \tag{22}$$

Success at least p thus requires both inequalities inside the maximum in Eq. (11). The proof allows arbitrary joint measurements and unlimited computational postprocessing. □

Declared register conventions and numerical meaning. Table 7 evaluates Eq. (11) at $p = 2/3$ for all listed models and four images. These are newly declared mathematical benchmark settings; they are not reconstructed experimental parameters. With $n = \lceil \log_2 N \rceil$, the feature counts used here are

Model	r
FRQI	1
NEQR	8
QIRHSI/EQIR	10
AFQIRHSI ($p_{\mathrm{AF}} = 4$)	$p_{\mathrm{AF}} + 3 = 7$
QIRBP ($b_{\mathrm{BP}} = 0$)	$b_{\mathrm{BP}} + 6 = 6$
QRMW (3 channels)	$8 + \lceil \log_2 3 \rceil = 10$
QIRWC ($b_{\mathrm{WC}} = 0$)	$n + 2b_{\mathrm{WC}} + 8 = n + 8.$

(23)

All selected feature qubits are included in the controlled substitution assumed by the theorem. In particular, QIRWC uses one main position address of size N; its other position register is counted among the feature qubits. Thus its total register count is $Q = 2n + 8$ and its valid main-address dimension is $D = N2^{n+8}$. It is not counted as a pair-address image with N^2 main addresses.

The stated QIRBP sum over positions, bit planes, and wavelength labels has normalization

$$(N2^{b_{\mathrm{BP}}+3})^{-1/2}.$$

The choice $b_{\mathrm{BP}} = 0$ therefore uses $r = 6$ in this table. The previous illustrative $r = 11$ calculation would correspond to $b_{\mathrm{BP}} = 5$ under this register convention; it does not establish that this was an experimental setting.

The table values are necessary ciphertext-preparation counts under the specified angle law and register conventions. They are neither measured attack times nor bit-security exponents. The logarithm of a sample count must not be labelled as elementary-gate security. Each displayed bound is rounded conservatively; the accompanying calculation data retain integer thresholds.

To relate the reference calculation to an actual generator, suppose the resulting average ciphertext is within trace distance κ of the reference ciphertext, uniformly over the allowed input images. For example, a bound κ on the total variation distance of the actual joint finite angle/control-key law from its finite reference law suffices. Comparing a finite law directly to a continuous law in total variation is inappropriate; the reference must use the same finite alphabet. Then the preceding proof gives

$$\begin{aligned} C_\kappa &= \min\left\{ \ln D,\ \ln\left(1 + D[\beta_L\sqrt{1 - 1/N} + 2\kappa]^2\right)\right\}, \\ q &\geq \left\lceil \max\left\{ \frac{\ln 2 - h(p)}{C_\kappa},\ \frac{p - 1/2}{\beta_L(1 - 1/N) + \kappa}\right\}\right\rceil. \end{aligned} \tag{24}$$

The trace-distance bound becomes $\frac{1}{2}\|\sigma - \tau\|_1 \leq \beta_L(1 - 1/N) + \kappa$. This follows from $\|\sigma - \sigma_{\mathrm{ref}}\|_2 \leq \|\sigma - \sigma_{\mathrm{ref}}\|_1 \leq 2\kappa$. The source bound must include correlations between pixels, axes, source parameters, both components of B, and any relevant side information. Statistical image metrics or balanced one-bit marginals do not certify this joint bound.

The mapping $\theta = \gamma \bmod (q, \pi)$ does not by itself establish the reference law. Even if $q \bmod \pi$ were uniform, its angle range would be $[0, \gamma\pi)$; the reference continuous range requires $\gamma = 2$, as well as the stated joint moment conditions. No measured value of κ or proof of those conditions for the original generators is supplied here. Therefore the numerical rows are conditional benchmarks and are not asserted as certified resistance of the experimental generator.

Why the range cannot be omitted. There is an explicit contrasting calculation within the same circuit family. Suppose the Euler angles are instead independent uniform variables on $[0, \pi)$, independently of B. Averaging the Bloch rotations gives, for a data qubit initially in $|0\rangle$ or $|1\rangle$,

$$\langle X\rangle_0 = (2/\pi)^2, \qquad \langle X\rangle_1 = -(2/\pi)^2. \tag{25}$$

Table 7: Conditional QEF image-discrimination sample bounds for all listed QIR models and the four original images. The original generator is not certified by these reference calculations.

(a) Ashurbanipal image, $H \times W = 256 \times 256$, $n = \lceil \log_2(HW) \rceil = 16$

Model	r	$Q = n + r$	$D = N2^r$	q_2	q_∞	$H_2 \geq$	Source
FRQI	1	17	$65{,}536\,2^1$	≥ 2	≥ 3	6	NC
NEQR	8	24	$65{,}536\,2^8$	$\geq 2.79624 \times 10^6$	$\geq 4.54809 \times 10^{10}$	48	NC
QIRHSI	10	26	$65{,}536\,2^{10}$	$\geq 1.98605 \times 10^9$	$\geq 5.77734 \times 10^{14}$	60	NC
EQIRHSI	10	26	$65{,}536\,2^{10}$	$\geq 1.98605 \times 10^9$	$\geq 5.77734 \times 10^{14}$	60	NC
AFQIRHSI	7	23	$65{,}536\,2^7$	$\geq 349{,}531$	$\geq 4.08705 \times 10^8$	42	NC
QIRBP	6	22	$65{,}536\,2^6$	$\geq 43{,}692$	$\geq 3.97626 \times 10^6$	36	NC
QRMW	10	26	$65{,}536\,2^{10}$	$\geq 1.98605 \times 10^9$	$\geq 5.77734 \times 10^{14}$	60	NC
QIRWC	$16 + 8$	40	$65{,}536\,2^{24}$	$\geq 2.29734 \times 10^{30}$	$\geq 3.12918 \times 10^{43}$	144	NC

(b) Hammurabi code image, $H \times W = 222 \times 227$, $n = \lceil \log_2(HW) \rceil = 16$

Model	r	$Q = n + r$	$D = N2^r$	q_2	q_∞	$H_2 \geq$	Source
FRQI	1	17	$50{,}394\,2^1$	≥ 2	≥ 3	6	NC
NEQR	8	24	$50{,}394\,2^8$	$\geq 3.04381 \times 10^6$	$\geq 5.91175 \times 10^{10}$	48	NC
QIRHSI	10	26	$50{,}394\,2^{10}$	$\geq 2.57085 \times 10^9$	$\geq 7.51324 \times 10^{14}$	60	NC
EQIRHSI	10	26	$50{,}394\,2^{10}$	$\geq 2.57085 \times 10^9$	$\geq 7.51324 \times 10^{14}$	60	NC
AFQIRHSI	7	23	$50{,}394\,2^7$	$\geq 349{,}533$	$\geq 5.29581 \times 10^8$	42	NC
QIRBP	6	22	$50{,}394\,2^6$	$\geq 43{,}692$	$\geq 5.05218 \times 10^6$	36	NC
QRMW	10	26	$50{,}394\,2^{10}$	$\geq 2.57085 \times 10^9$	$\geq 7.51324 \times 10^{14}$	60	NC
QIRWC	$16 + 8$	40	$50{,}394\,2^{24}$	$\geq 2.98764 \times 10^{30}$	$\geq 4.06943 \times 10^{43}$	144	NC

(c) Lamassu image, $H \times W = 256 \times 256$, $n = \lceil \log_2(HW) \rceil = 16$

Model	r	$Q = n + r$	$D = N2^r$	q_2	q_∞	$H_2 \geq$	Source
FRQI	1	17	$65{,}536\,2^1$	≥ 2	≥ 3	6	NC
NEQR	8	24	$65{,}536\,2^8$	$\geq 2.79624 \times 10^6$	$\geq 4.54809 \times 10^{10}$	48	NC
QIRHSI	10	26	$65{,}536\,2^{10}$	$\geq 1.98605 \times 10^9$	$\geq 5.77734 \times 10^{14}$	60	NC
EQIRHSI	10	26	$65{,}536\,2^{10}$	$\geq 1.98605 \times 10^9$	$\geq 5.77734 \times 10^{14}$	60	NC
AFQIRHSI	7	23	$65{,}536\,2^7$	$\geq 349{,}531$	$\geq 4.08705 \times 10^8$	42	NC
QIRBP	6	22	$65{,}536\,2^6$	$\geq 43{,}692$	$\geq 3.97626 \times 10^6$	36	NC
QRMW	10	26	$65{,}536\,2^{10}$	$\geq 1.98605 \times 10^9$	$\geq 5.77734 \times 10^{14}$	60	NC
QIRWC	$16 + 8$	40	$65{,}536\,2^{24}$	$\geq 2.29734 \times 10^{30}$	$\geq 3.12918 \times 10^{43}$	144	NC

(d) Babylon Lion image, $H \times W = 768 \times 1024$, $n = \lceil \log_2(HW) \rceil = 20$

Model	r	$Q = n + r$	$D = N2^r$	q_2	q_∞	$H_2 \geq$	Source
FRQI	1	21	$786{,}432\,2^1$	≥ 2	≥ 3	6	NC
NEQR	8	28	$786{,}432\,2^8$	$\geq 2.79620 \times 10^6$	$\geq 3.87854 \times 10^9$	48	NC
QIRHSI	10	30	$786{,}432\,2^{10}$	$\geq 1.98872 \times 10^8$	$\geq 4.81641 \times 10^{13}$	60	NC
EQIRHSI	10	30	$786{,}432\,2^{10}$	$\geq 1.98872 \times 10^8$	$\geq 4.81641 \times 10^{13}$	60	NC
AFQIRHSI	7	27	$786{,}432\,2^7$	$\geq 349{,}526$	$\geq 3.95317 \times 10^7$	42	NC
QIRBP	6	26	$786{,}432\,2^6$	$\geq 43{,}691$	$\geq 1.91498 \times 10^6$	36	NC
QRMW	10	30	$786{,}432\,2^{10}$	$\geq 1.98872 \times 10^8$	$\geq 4.81641 \times 10^{13}$	60	NC
QIRWC	$20 + 8$	48	$786{,}432\,2^{28}$	$\geq 2.00742 \times 10^{35}$	$\geq 4.22549 \times 10^{50}$	168	NC

q_2 and q_∞ are necessary independent ciphertext preparations for success $2/3$, from Eq. (11); the latter is an ideal continuous reference, not a finite-key claim. H_2 is the necessary conditional source entropy (bits) for the $L = 2$ independent-tuple reference, not a security exponent. NC means the original source conditions are not certified. The declared benchmark uses 8-bit intensity, AFQIRHSI $p = 4$, RGB QRMW, and one wavelength label ($b = 0$) for QIRBP/QIRWC. In QIRWC the second position register is included among the features, so $r = n + 8$. These are explicit mathematical settings, not reconstructed experiment metadata. Valid dimension D differs from padded $d = 2^Q$ when $N < 2^n$. Large lower bounds are rounded down; exact integer ceilings are supplied separately.

Consider all-black and all-white images with a feature data qubit initially $|0\rangle$ and $|1\rangle$, respectively. At output position $t = 0$, no position control is active, so $V_B(0) = I$. That position occurs with probability $1/N$, regardless of the permutation. An adversary can request fresh encryptions until measuring $t = 0$, then measure X on that feature qubit. This stopping-time procedure uses N preparations on average and succeeds with probability

$$\frac{1 + (2/\pi)^2}{2} \simeq 0.702642 > 2/3. \tag{26}$$

For a fixed budget of q preparations, use the first such outcome and guess uniformly if none is obtained. The resulting success is

$$P_{\text{attack}}(q) = \frac{1}{2} + \frac{2}{\pi^2}[1 - (1 - 1/N)^q]. \tag{27}$$

Therefore this contrasting law admits a fixed-budget attack with success at least $2/3$ at

$$q_{\text{attack}} = \left\lceil \frac{\ln(1 - \pi^2/12)}{\ln(1 - 1/N)} \right\rceil. \tag{28}$$

For the four images in table order these counts are $113,285$, $87,111$, $113,285$, and $1,359,425$. They are attack upper bounds under the contrasting law, distinct from the theorem's lower bounds under its reference law. This is not a claimed attack on an unspecified nD-QHCS distribution. It shows why the source range and joint moments cannot be omitted from a numerical security claim.

Certification of the original angle source. The reference law above must be distinguished from a property of the implemented generator. Write $Z = q \bmod \pi$. If q has density f_q, then, for $0 \le z < \pi$,

$$f_Z(z) = \sum_{m \in \mathbb{Z}} f_q(z + m\pi), \qquad f_\theta(t) = \gamma^{-1} f_Z(t/\gamma), \quad 0 \le t < \gamma\pi. \tag{29}$$

Indeed, the inverse image of a measurable set $A \subset [0, \pi)$ is the disjoint union of $A + m\pi$; summing their probabilities and changing variables proves the first identity. Scaling proves the second. Thus $\gamma = 2$ and $f_Z = 1/\pi$ are sufficient for the one-pass continuous marginal reference on $[0, 2\pi)$, but the modulo operation does not establish either uniformity or independence. The manuscript does not specify an actual numerical γ, a probability law for the initial source state, or a sampling map that proves the required conditional pixel/axis moments. Chaotic sensitivity and a bounded angle range do not provide that missing probability law.

A bound stated directly in terms of source moments. Full independence is sufficient for the preceding reference calculation, but is not necessary for a security bound. Let $F = 2^r$, $D = NF$, and condition on $B = b$ in its essential support. For the actual substitution operators define

$$\delta_2 = \operatorname*{ess\,sup}_{b,k,\|u\|=1} \left\| \mathbb{E}_{A|b}[R_{A,k}|u\rangle\langle u|R_{A,k}^\dagger] - I_F/F \right\|_2, \tag{30}$$

$$\eta_2 = \operatorname*{ess\,sup}_{b,k\neq l,\|u\|=\|v\|=1} \left\| \mathbb{E}_{A|b}[R_{A,k}|u\rangle\langle v|R_{A,l}^\dagger] \right\|_2. \tag{31}$$

Here vectors have Euclidean norm one and $\|\cdot\|_2$ denotes the Hilbert–Schmidt matrix norm. Public setup information is also conditioned on. These quantities account for dependence between the two source classes, between axes, and between pixel tuples.

Theorem 2 (Source-moment certification criterion)**.** *For image states in Eq.* (1) *and the original clean three-layer unitary, the actual average ciphertext satisfies*

$$\mathsf{D}(\sigma \| I_D/D) \le C_{\text{actual}}, \qquad C_{\text{actual}} = \ln\left[1 + F\delta_2^2 + (N-1)F\eta_2^2\right]. \tag{32}$$

If whole key records are independent between preparations, success $p > 1/2$ in the binary image experiment requires

$$\boxed{q \ge \left\lceil \frac{\ln 2 - h(p)}{C_{\text{actual}}} \right\rceil}. \tag{33}$$

When $C_{\text{actual}} = 0$, no finite number of these preparations distinguishes the candidate images.

Proof. Before applying $E_b P_b$, subtract $\tau = I_D/D$ from the conditional average. Its diagonal position blocks are $1/N$ times the matrices in Eq. (30); its off-diagonal blocks are $1/N$ times those in Eq. (31). Distinct blocks are orthogonal in the Hilbert–Schmidt inner product. Unitary conjugation preserves this norm and τ; convexity of its square therefore gives

$$\|\sigma - \tau\|_2^2 \le \delta_2^2/N + (1 - 1/N)\eta_2^2. \tag{34}$$

Using $\mathrm{D}(\sigma\|\tau) \le \ln(1 + D\|\sigma - \tau\|_2^2)$ proves Eq. (32). The same Holevo/Fano argument used above proves Eq. (33). No independence within an individual key record is used in this proof. □

The reference law has $\delta_2 = 0$ and $\eta_2 = \beta_L$. For an uncertified source, only the general inequalities $\delta_2^2 \le 1 - 1/F$ and $\eta_2 \le 1$ are available. They give $C_{\text{actual}} \le \ln D$ and only the trivial necessary count $q \ge 1$ for $p = 2/3$ and $D \ge 2$. This is not a successful attack; it means the source information supplied does not certify a larger count. A nontrivial value must follow from certified upper bounds on the actual conditional moments, rather than being assigned from the reference table.

Finite keys and the stronger independence assumption. Two independent uniform pixel tuples over an L-point angle alphabet contain $6r$ independent symbols. Consequently

$$H(A \mid B) \ge H(\Theta_k, \Theta_l \mid B) = 6r \log_2 L, \qquad k \ne l. \tag{35}$$

If all angles are deterministic expansions of an s-bit substitution seed, their pair has at most 2^s possible values. For its support S, the actual law gives probability one to S, whereas the independent uniform pair law gives at most $2^s/L^{6r}$. Hence their total variation distance is at least

$$\max\{0, 1 - 2^s/L^{6r}\}. \tag{36}$$

For QIRBP ($r = 11$) and a deterministic 128-bit substitution seed, this lower bound is $15/16$ at $L = 4$, and $1 - 2^{-400}$ at $L = 256$. These results concern the strong independent-symbol reference; they do not exclude a correlated source satisfying the weaker operator-moment criterion in Theorem 2.

What the HGΦ measurement establishes. For the stated raw-bit model, an observer who knows the phase and reference setting has single-bit guessing probability and conditional pointwise min-entropy

$$p_{\text{guess}}(b \mid \phi, \delta) = \frac{1 + |\cos(\phi + \delta)|}{2}, \qquad H_{\min}(b \mid \phi, \delta) = -\log_2 p_{\text{guess}}(b \mid \phi, \delta). \tag{37}$$

The latter can be zero. A fixed deterministic coarse-graining or hash cannot create missing entropy; a reversible map preserves it. Specifying a randomness extractor would additionally require a source entropy bound, seed assumptions, output length, and extraction error. None is supplied here as a certification of the original generator.

Freshness alone does not supply the missing bound. An explicit countermodel uses two constant FRQI images with feature states $|0\rangle$ and $|1\rangle$. Set the x and y rotation angles to zero, refresh the z angles independently for every encryption, allow any position permutation, and let the entanglement component select CZ gates. Every layer preserves the feature's Z basis value. One feature measurement therefore distinguishes the two images with probability one using a single ciphertext. The whole key record is fresh even though its gate-selection component is fixed. This is an allowed source-law countermodel to an architecture-only claim, not an attack attributed to the unspecified actual nD-QHCS/HGΦ law.

Finally, the supplied experimental implementation resets NumPy's scalar-seeded pseudorandom generator and uses a fixed seed in its driver. Its QIRBP branch performs shared channel masks and XOR chaining; it does not implement the controlled complex Euler rotations or the specified nD-QHCS/HGΦ generator. It therefore cannot certify the moments in Eqs. (30)– (31). The actual-source numerical resistance remains unestablished; the preceding table entries are conditional mathematical benchmarks.

Interpretation of the original attack labels. Independent known/chosen-image pairs from other operations do not share the target key under the independent-refresh assumption. They can still reveal a biased generator law or shared source parameters; the conditional key law must therefore be specified. Generic key search, if a reversible candidate verifier is available, has the usual conditional $\Theta(\sqrt{M/t})$ query baseline for M candidates and t marked candidates [2, 3, 4]. No particular old key-length value is used to certify QEF, and no reduction proving that all attacks require exhaustive search is claimed. The dimension or parameter count of a general quantum state or unitary is not substituted for an attack lower bound.

Theorem 1 concerns the mathematical quantum operators and the stated key law. Classical array simulations and their rendered images do not verify these quantum moment conditions. In particular, the original scalar-seeded XOR/permutation simulation is not an implementation of the controlled complex Euler rotations assumed here. Image-quality and decryption metrics are therefore reported as diagnostics, rather than as evidence that the conditional security bound holds. The new probabilistic bound is proved analytically; it is not included in the existing Lean 4 unitarity checks. Entanglement alone supplies neither authentication nor a proof of resistance to active modification.

2 Verification and effects of the corrections

The numerical supplement has 128 rows: eight allocations, four native image sizes, and four angle reference laws ($L = 2, 4, 256$ and the ideal continuous limit). Integer thresholds agree at 120 and 170 decimal digits. The exact entropy denominator and the trace-distance converse strengthen the earlier purity-based bound; they do not change encryption.

The main manuscript corrects both PSB phase signs and proves gauge invariance. This affects the phase, Bernoulli probability, and any keys generated from that probability; old source records need recomputation. The original source classes are retained. The security table keeps four panels and tan headers, separates QRMW and QIRWC, and declares each register allocation. QIRBP's plane-index qubits are counted separately from intensity precision. The QIRWC secondary address is included in its feature register.

The experimental scope, statistical interpretations, source indexing, resource counts, ancillary cleanup discussion, and numerical discussion are adjusted consistently. Existing image metrics remain diagnostics of the supplied array simulations. The source moment bound is an analytic result, not part of the earlier Lean unitarity checks. The original manuscript and the previous review copy are retained.

Independent verification of the strengthened bound. Four finite reference models were checked by exhaustive substitution-key enumeration (64, 512, 4096, and 4096 records), including complete superoperators, first moments, and feature twirls. Exact spectra and relative entropies were checked on 48 normalized image states. Maximum superoperator, spectrum, and entropy errors were 1.8e-16, 2.5e-16, and 5.83e-16, respectively. Optimal joint binary measurements in 10 reference cases satisfied both converses. Two eight-record sources with correlated substitution and control material were checked on 40 image states (block-orthogonality error 1.11e-16), with 5 additional joint-measurement checks. Eight source-perturbation cases checked the robustness inequalities. All 128 table rows, dimensions, and integer ceilings were independently recomputed at 145 decimal digits using a different pi algorithm and a cancellation-resistant entropy formula. The fixed-budget range attack and 48 fresh-record Z/CZ outcomes were also checked. These calculations support the derivation; they do not certify the original source law or report cryptanalytic attacks or hardware performance.

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